

TITAN EARTH GRID® 20

BI-AXIAL POLYPROPYLENE GEOGRID



Titan Earth Grid® 20 is a bi-axial polypropylene (PP) geogrid. It is manufactured using a punching and drawing process whereby the polypropylene sheet is stretched in two directions, machine (longitudinal) and cross-machine (transverse). The result is a monolithic and isotropic geogrid with thick and wide ribs, and thick integral nodes. The ribs have a high degree of molecular orientation continuing in part through the mass of the integral node. Engineered to be mechanically and chemically stable in aggressive soil environments, the Earth Grid™ geometry allows for strong mechanical interlock with soil particles and features high tensile stiffness at low strains to resist construction damage, environmental exposure and formulated to resist UV degradation. It is also not susceptible to hydrolysis, environmental stress cracking and micro-organism attack.

TESTED PROPERTY	TEST METHOD	UNIT ENGLISH (METRIC)	VALUE ENGLISH (METRIC)	
			MD	XD
Ultimate Tensile Strength ⁽¹⁾	ASTM D 6637	lbs/ft (kN/m)	1,371 (20.0)	1,371 (20.0)
Tensile Strength at 2% Strain ⁽¹⁾	ASTM D 6637	lbs/ft (kN/m)	548 (8.0)	548 (8.0)
Tensile Strength at 5% Strain ⁽¹⁾	ASTM D 6637	lbs/ft (kN/m)	960 (14.0)	960 (14.0)
Minimum Carbon Black Content	ASTM D 4218	%	2	
Junction Strength ⁽¹⁾⁽³⁾	GRI-GG2 ASTM D 7737	lbs/ft (kN/m)	1,302 (19.0)	1,302 (19.0)
Flexural Rigidity ⁽¹⁾	ASTM D 7748	mg-cm	750,000	
Aperture Stability ⁽²⁾⁽⁴⁾	US. COE	m-N/deg	0.50	
Resistance to UV degradation @ 500 hours ⁽²⁾	ASTM D 4355	%	100	
Minimum Rib Thickness	Callipered	inch (mm)	0.06 (1.4)	0.04 (1.0)
Aperture Size ⁽²⁾⁽⁵⁾	Nominal	inch (mm)	1.38 (35.0)	1.38 (35.0)
TYPICAL ROLL DIMENSIONS				
Roll Width		ft (m)	12.95 (3.95)	
Roll Length ⁽⁶⁾		ft (m)	164.04 (50.0)	

NOTES:

(1) Minimum Average Roll Values (MARV) – Calculated as (mean minus 2x standard deviation) – ASTM-D4759-02.

(2) Average.

(3) Junction efficiency is defined as junction strength divided by multi-rib strength.

(4) Resistance to in plane rotational movement measure at an applied moment = 2m-N (20kg-cm) in accordance with US Army Corps of Engineers methodology for the measurement of torsional rigidity.

(5) Aperture tolerance: within ± 10% coefficient of variance.

(6) Custom Length orders can be accommodated.

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